Dallas International School

Test 4

Topics 1.5-1.7-2.2-2.3-2.4

11th IB - Math AA

For SL & HL Students

Name : . . . . . .

EXERCISE 1 The temperature of a cup of tea, t minutes after it is poured, can be modelled by

$$H(t) = 20 + 75e^{-0.4t}, \quad t \geq 0.$$

The temperature is measured in degrees Celsius (°C).

1. a. Find the initial temperature of the tea. [1]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

b. Find the temperature of the tea 1h10 after it is poured. [1]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

2. After k minutes, the tea will be below 35°C and cool enough to drink.

Find the least possible value of k, where $k \in \mathbb{Z}^+$. [3]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

As the tea cools, $H(t)$ approaches the temperature of the room, which is constant.

3. Find the temperature of the room. [2]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

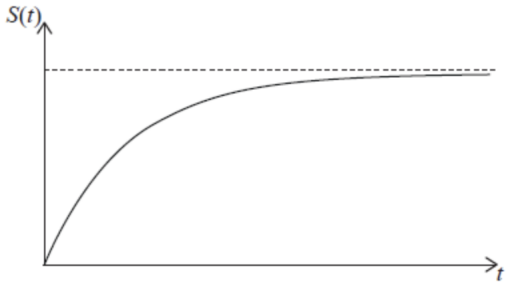
EXERCISE 2

Jean-Pierre jumps out of an airplane that is flying at constant altitude. Before opening his parachute, he goes through a period of freefall.

Jean-Pierre's vertical speed during the time of freefall, S, in m s^{-1}, is modelled by the following function :

$$S(t) = K - 40\left(1.6^{-5t}\right), \quad t \geq 0$$

where t is the number of seconds after he jumps out of the airplane, and K is a constant. A sketch of Jean-Pierre's vertical speed against time is shown below.



Jean-Pierre's initial vertical speed is 0 m s^{-1}.

1. Find the value of K. [2]

. . . . . .

. . . . . .

. . . . . .

2. In the context of the model, state what the horizontal asymptote represents. [1]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

3. Find Jean-Pierre's vertical speed after 20 seconds. Give your answer in km h^{-1}. [3]

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

EXERCISE 3

The functions f and g are defined such that

$$f(x) = \frac{x+3}{4} \qquad \text{and} \qquad g(x) = 8x + 5.$$

1. Show that $(g \circ f)(x) = 2x + 11$. [2]

2. Given that $(g \circ f)^{-1}(a) = 4$, find the value of a. [3]

Answer Key

1. **a.** $H(0) = 20 + 75e^0 = 95°\text{C}$.

b. $1h10 = 70$ minutes :

$$H(70) = 20 + 75e^{-0.4\times 70} \approx 20°\text{C}.$$

2. Using the GDC and intersect with $y = 35$ $\boxed{k \approx 4.02 \quad \text{so} \quad 5}$.

3.

$$\lim_{t\to+\infty} H(t) = 20.$$

So the room temperature is $\boxed{20°\text{C}}$.

1.

$$0 = K - 40(1.6^0) = K - 40 \quad \Longrightarrow \quad \boxed{K = 40}.$$

2. Horizontal asymptote : the terminal (maximum) vertical speed during freefall, equal to $\boxed{40 \text{ m s}^{-1}}$.

3.

$$S(20) = 40 - 40\left(1.6^{-5\times 20}\right) = 40 - 40\left(1.6^{-100}\right) \approx 40 \text{ m s}^{-1}.$$

In km h^{-1} :

$$40 \times 3.6 = \boxed{144 \text{ km h}^{-1}}.$$

1.

$$(g \circ f)(x) = g\left(\frac{x+3}{4}\right) = 8\left(\frac{x+3}{4}\right) + 5 = 2(x+3) + 5 = \boxed{2x + 11}.$$

2. If $(g \circ f)^{-1}(a) = 4$, then $a = (g \circ f)(4)$:

$$a = 2 \cdot 4 + 11 = \boxed{19}.$$